

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF NETWORKING AND COMMUNICATIONS
21CSP401/21CSP402L – MAJOR PROJECT
LOG BOOK

Batch Details: NWC26B178

Regno: RA2211029010013, RA2211029010019

Name: Vedant Pandey, Chiranjeev Kumar

Project Title: Automated Network Topology Discovery And Analysis Using Cisco DNA Center

SPRINT 1 | Nov 01, 2025 – Jan 02, 2026 | Planning, Research & Initial Development

Date	Day	Team Member	What did I do Yesterday?	What will I do Today?	Blockers?	Progress	Notes
01-Nov-2025	Saturday	Vedant Pandey	Sprint planning	Problem definition & scope finalization	None	0%	Sprint 1 - Planning Kickoff
03-Nov-2025	Monday	Chiranjeev Kumar	Problem definition done	Literature survey on topology discovery	None	5%	Sprint 1 - Research
05-Nov-2025	Wednesday	Vedant Pandey	Literature survey	Finalize research gap & objectives	None	10%	Sprint 1 - Research
06-Nov-2025	Thursday	Chiranjeev Kumar	Research gap identified	Literature survey on topology methods	None	12%	Sprint 1 - Survey
10-Nov-2025	Monday	Vedant Pandey	Literature survey completed	Identify research gap & objectives	None	15%	Sprint 1 - Gap Analysis
11-Nov-2025	Tuesday	Chiranjeev Kumar	Objectives finalized	Tool selection: Python, NetworkX, Scapy, SNMP	None	18%	Sprint 1 - Tools
15-Nov-2025	Saturday	Vedant Pandey	Tools selected	Initial system design & architecture planning	None	22%	Sprint 1 - Design
17-Nov-2025	Monday	Chiranjeev Kumar	Architecture draft ready	Review system design with supervisor	None	25%	Sprint 1 - Design Review

Date	Day	Team Member	What did I do Yesterday?	What will I do Today?	Blockers?	Progress	Notes
20-Nov-2025	Thursday	Vedant Pandey	Design finalized	System architecture planning	None	28%	Sprint 1 - Architecture
24-Nov-2025	Monday	Chiranjeev Kumar	Architecture done	Setup development environment & libraries	None	30%	Sprint 1 - Setup
26-Nov-2025	Wednesday	Vedant Pandey	Environment setup done	Cisco DNAC sandbox configuration	DNAC API auth challenges	35%	Sprint 1 - DNAC Config
01-Dec-2025	Monday	Chiranjeev Kumar	DNAC sandbox configured	API testing on DevNet Sandbox	DNAC API auth challenges	40%	Sprint 1 - API Test
05-Dec-2025	Friday	Vedant Pandey	API testing done	Implement DNAC discovery module	None	45%	Sprint 1 - DNAC Module
08-Dec-2025	Monday	Chiranjeev Kumar	DNAC module in progress	NetworkX graph model design & testing	None	50%	Sprint 1 - Graph Model
11-Dec-2025	Thursday	Vedant Pandey	Graph model designed	Implement DNAC discovery module	None	55%	Sprint 1 - DNAC Impl
15-Dec-2025	Monday	Chiranjeev Kumar	NetworkX graph model tested	GNS3 virtual lab setup	SNMP config setup needed	60%	Sprint 1 - GNS3 Setup
18-Dec-2025	Thursday	Vedant Pandey	GNS3 setup in progress	SNMP configuration in GNS3	SNMP config setup needed	65%	Sprint 1 - SNMP Config
22-Dec-2025	Monday	Chiranjeev Kumar	SNMP configured	GNS3 virtual lab finalization	None	70%	Sprint 1 - GNS3 Final
25-Dec-2025	Thursday	Vedant Pandey	GNS3 lab ready	SNMP/LLDP discovery module implementation	None	80%	Sprint 1 - SNMP/LLDP Module
29-Dec-2025	Monday	Chiranjeev Kumar	SNMP/LLDP module done	Sprint review & documentation	None	90%	Sprint 1 - Review
02-Jan-2026	Friday	Vedant Pandey	Sprint 1 review done	Prepare for Sprint 2	None	100%	Sprint 1 - Done

SPRINT 2 | Jan 05, 2026 – Mar 02, 2026 | Metrics, Analysis, Visualization & Integration

Date	Day	Team Member	What did I do Yesterday?	What will I do Today?	Blockers?	Progress	Notes
05-Jan-2026	Monday	Chiranjeev Kumar	Sprint 1 completed	Metric computation: NDR, LDR, MRS, Time	None	5%	Sprint 2 - Metrics
07-Jan-2026	Wednesday	Vedant Pandey	Metric framework designed	Implement NDR & LDR computation	None	10%	Sprint 2 - NDR/LDR
09-Jan-2026	Friday	Chiranjeev Kumar	NDR/LDR implemented	Metadata Richness Score computation	None	15%	Sprint 2 - MRS
12-Jan-2026	Monday	Vedant Pandey	MRS metric done	Discovery time measurement module	None	20%	Sprint 2 - Timing
16-Jan-2026	Friday	Chiranjeev Kumar	All metrics implemented	Comparative analysis: DNAC vs SNMP vs ARP	None	25%	Sprint 2 - Comparative
02-Feb-2026	Monday	Vedant Pandey	Comparative analysis started	ANOVA test for statistical validation	None	30%	Sprint 2 - ANOVA
04-Feb-2026	Wednesday	Chiranjeev Kumar	ANOVA test done	Pairwise t-tests with Bonferroni correction	None	40%	Sprint 2 - t-tests
06-Feb-2026	Friday	Vedant Pandey	Statistical validation complete	Comparative analysis implementation	None	50%	Sprint 2 - Results
09-Feb-2026	Monday	Chiranjeev Kumar	Analysis complete	Visualization module: charts & topology graphs	None	55%	Sprint 2 - Charts
11-Feb-2026	Wednesday	Vedant Pandey	Charts developed	Topology graph visualization	None	60%	Sprint 2 - Topology
16-Feb-2026	Monday	Chiranjeev Kumar	Visualization graphs done	Integration of viz module with backend	None	65%	Sprint 2 - Backend
18-Feb-2026	Wednesday	Vedant Pandey	Backend integration done	Interactive dashboard (Flask routes & UI)	None	70%	Sprint 2 - Dashboard
23-Feb-2026	Monday	Chiranjeev Kumar	Flask routes linked	Dashboard UI improvements	None	75%	Sprint 2 - UI
25-Feb-2026	Wednesday	Vedant Pandey	Dashboard ready	Export manager: JSON & CSV formats	None	80%	Sprint 2 - Export
27-Feb-2026	Friday	Chiranjeev Kumar	Export manager done	Sprint 2 review & testing	None	90%	Sprint 2 - Review
02-Mar-2026	Monday	Vedant Pandey	Sprint 2 reviewed	Prepare Sprint 3	None	100%	Sprint 2 - Done

SPRINT 3 | Mar 02, 2026 – Apr 30, 2026 | Testing, Optimization & Final Deployment

Date	Day	Team Member	What did I do Yesterday?	What will I do Today?	Blockers?	Progress	Notes
02-Mar-2026	Monday	Chiranjeev Kumar	Sprint 2 completed	Anomaly detection: proxy ARP & rogue nodes	None	5%	Sprint 3 - Anomaly
04-Mar-2026	Wednesday	Vedant Pandey	Anomaly detection started	Testing proxy ARP detection heuristic	None	10%	Sprint 3 - Proxy ARP
06-Mar-2026	Friday	Chiranjeev Kumar	Proxy ARP detection tested	Decision engine testing with profiles	None	15%	Sprint 3 - Decision Engine
09-Mar-2026	Monday	Vedant Pandey	Decision engine tested	Validate decision engine across network profiles	None	25%	Sprint 3 - Validation
11-Mar-2026	Wednesday	Chiranjeev Kumar	Validation done	LLDP validation checks implementation	None	35%	Sprint 3 - LLDP Validation
16-Mar-2026	Monday	Vedant Pandey	LLDP checks added	SNMP parallel processing optimization	None	45%	Sprint 3 - SNMP Opt.
23-Mar-2026	Monday	Vedant Pandey	Performance improved	Bug fixing across modules	None	65%	Sprint 3 - Bug Fix
06-Apr-2026	Monday	Chiranjeev Kumar	Bugs fixed	Integration of visualization with backend graph	None	70%	Sprint 3 - Final Viz
09-Apr-2026	Thursday	Vedant Pandey	Viz integration done	Interactive dashboard Flask routes & UI linking	None	75%	Sprint 3 - Dashboard
18-Apr-2026	Saturday	Vedant Pandey	Export manager done	Anomaly detection module final testing	None	85%	Sprint 3 - Anomaly Test
21-Apr-2026	Tuesday	Chiranjeev Kumar	Anomaly testing done	Decision engine final validation	Minor UI improvements needed	90%	Sprint 3 - Engine Final
24-Apr-2026	Friday	Vedant Pandey	Engine validated	Performance optimization & final bug fix	Minor UI improvements needed	95%	Sprint 3 - Optimize
28-Apr-2026	Tuesday	Chiranjeev Kumar	All modules finalized	Final documentation & project submission	None	98%	Sprint 3 - Docs
30-Apr-2026	Thursday	Vedant Pandey	Documentation ready	Final project review & submission	None	100%	Sprint 3 - Completed